Kahan's method on quadratic lifts: modified invariants, Liouville measures and integrable maps in every dimension

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Abstract

Kahan's method discretises a quadratic vector field by a birational map that costs one linear solve per step and often inherits invariant measures, modified integrals and integrability. Quadratic lifts (auxiliary variables such as $y = x_i x_j$ or $u = \mathrm{e}^q$ that make a polynomial or exponential vector field quadratic) carry the method beyond quadratic problems, but lift constraints and original invariants need not survive. We show which do. (i) A *block theorem*: variables that move along fixed affine flows at affine speeds keep modified versions of the quadratic invariants of those flows when their second-derivative operators close and commute; the flows themselves need not commute. It contains the two-variable theorem of Celledoni et al., gives invariants with squared denominators for resonant blocks, and gives $I - \frac{h^2}{8} F^\top \nabla^2 I \, F$ when the curvature of I lies in affinely moving variables. (ii) Constraints with affinely moving factors are protected, and on the invariant *shifted sheet* the lifted method is a symmetric one-step method on the original variables. (iii) For potentials built from exponentials and single harmonics, with any mass matrix, the lifted map preserves the Liouville measure and a modified energy. (iv) For isotropic quartic oscillators in $\mathbb{R}^n$ the lifted map is integrable for every n: we give its modified energy, angular momenta and measure in closed form with elementary proofs, show that compact regular level sets are invariant tori, and obtain an $O(n)$-cost integrator that, at moderate amplitudes, stays stable far beyond the Störmer–Verlet limit. Degree-growth computations suggest that integrability survives lifting when it stems from linear symmetries and one degree of freedom.

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1 Introduction

Let $\dot{z} = F(z)$ be a quadratic vector field on $\mathbb{R}^N$, $F(z) = Q(z,z) + Bz + b$ with Q symmetric and bilinear. Kahan's method [1, 2] replaces it by the map $z \mapsto z'$ defined by

$$\frac{z' - z}{h} = Q(z, z') + \tfrac{1}{2} B(z + z') + b. \tag{1}$$

The new point enters linearly, so a step costs one linear solve. The method is symmetric and, when applied to Hamiltonian quadratic vector fields (cubic Hamiltonians) with constant Poisson structure, it preserves a modified energy and an invariant measure [7]. Applied to integrable systems it produces integrable maps far more often than one has a right to expect: the Euler and Lagrange tops [3, 4], the Clebsch and Kirchhoff systems and many others [5, 6, 8, 12, 13], planar cubic Hamiltonian systems [13, 15], and cases where integrability can be restored by small

adjustments of the coefficients [14]. Kahan's method is therefore one of the few places where geometric numerical integration [25] and the theory of discrete integrable systems [26, 27] meet.

The method is defined only for quadratic vector fields. Polarisation extends it to higher degree at the price of multistep schemes [9], and Hone and Quispel construct explicit analogues for certain higher order equations [16]. Another route is a *quadratic lift*. Introduce auxiliary variables, for instance $y_{ij} = x_i x_j$, $u = \mathrm{e}^{\alpha \cdot q}$ or $(c, s) = (\cos\theta, \sin\theta)$, so that the original system becomes the restriction of a quadratic system in more variables to an invariant "sheet". Every polynomial vector field admits such a lift [18], and lifts are widely used in model reduction [19, 20, 21]. In geometric integration, auxiliary variables make energy-stable schemes linear (the IEQ and SAV approaches [22, 23]), and multiple quadratic auxiliary variables allow the midpoint rule to preserve several invariants at once [24]. Combining a quadratic lift with Kahan's method gives a linearly implicit one-step integrator for any polynomial ODE, and for many non-polynomial ones.

The price is that Kahan's map lives in the lifted space. It need not preserve the sheet, and the integrals and measures of the original system need not survive. This paper asks which structures do survive, and whether lifting can produce new integrable maps. Our main results are the following.

(i) **A block theorem** (Theorem 3.2). If a block of variables moves as $\dot{w} = \sum_k A_k(z) L_k(w)$, with affine *speeds* A_k and fixed affine vector fields L_k, then Kahan's map conserves explicitly modified versions of all quadratic invariants of the L_k, provided the second-derivative operators of the L_k close on these invariants and commute. The fields L_k themselves need not commute. The theorem contains the two-variable theorem of [10], gives modified invariants with squared denominators for resonant blocks (Example 3.6), and yields $I_h = I - \frac{h^2}{8} F^\top \nabla^2 I \, F$ whenever the curvature of a quadratic integral I lies in variables that move affinely (Corollary 3.4).

(ii) **Lift constraints and the shifted sheet** (Section 4). We classify the lift constraints $y - x_i x_j$ that are exact integrals (Proposition 4.1). Those whose factors move affinely are always protected, as $y - x_i x_j + \frac{h^2}{4} \dot{x}_i \dot{x}_j$ (Theorem 4.2). On the invariant "shifted sheet" that these modified constraints define, the lifted method is a symmetric second-order one-step method on the original variables (Theorem 4.4).

(iii) **Liouville measures** (Theorem 5.1). For every Hamiltonian $\frac{1}{2} p^\top G p + \sum_k c_k \mathrm{e}^{\alpha_k \cdot q} + \sum_j g_j \cos(\beta_j \cdot q)$, in any dimension and with any constant mass matrix, Kahan's method in the exponential and harmonic variables preserves the canonical Liouville measure and a modified energy. This covers Toda chains, rotor and XY chains and coupled pendula.

(iv) **An integrable family in every dimension** (Theorem 6.1 and Corollary 6.2). For $H = \frac{1}{2}|p|^2 + U(|q|^2)$ with U quadratic and $q, p \in \mathbb{R}^n$, the lift $y = |q|^2$ gives a Kahan map that conserves modified angular momenta, a modified energy and an explicit measure, for every n. Its compact regular level sets are invariant 2-tori on which it acts as a skew rotation. The map can be evaluated in closed form at $O(n)$ cost (Remark 6.4). Bounded orbits are guaranteed under an explicit condition (Proposition 6.3), and in our tests at moderate amplitudes the map stays stable far beyond the Störmer–Verlet limit (Section 8).

(v) **Which integrable systems survive lifting** (Section 7). Exact degree-growth computations on nineteen lifted maps indicate that the lifted Kahan map is integrable when the original integrability comes from linear symmetries plus one degree of freedom, and is not when it rests on hidden symmetries (Conjecture 7.1).

Relation to other work. A companion manuscript [17] studies Kahan's method for Lie–Poisson systems through a Cayley form of the map. Its "midpoint theorem" covers the single-speed case of our block theorem and the commuting cases of our measure results. Our block theorem

does not need commuting coefficient matrices, which is what the isotropic family requires (Remark 3.8). Conversely, [17] proves, with computer assistance, that every three-dimensional quadratic Nambu system has a Kahan-invariant measure, which we use in Section 5.2. Integrable Kahan maps in arbitrary dimension are known for Lotka–Volterra systems [37, 38] and for almost abelian Lie–Poisson systems [17]; the family of Section 6 is of a different kind, coming from a mechanical system with a non-quadratic potential through a lift. Integrable discretisations of central-force and quartic-potential problems have been constructed by other means [28, 29, 27], and discrete-gradient methods preserve energy and angular momentum [30]. What is new in Section 6 is that integrability arises from a general-purpose, linearly implicit integrator, with all modified integrals and the measure in closed form and in every dimension.

Outline. Section 2 collects facts about Kahan's method and defines quadratic lifts. Section 3 proves the block theorem and its corollaries. Section 4 treats lift constraints and the shifted sheet, Section 5 invariant measures, and Section 6 the isotropic family. Section 7 surveys integrability, Section 8 gives numerical experiments, and Section 9 lists open problems. Appendix A describes an independent computer-algebra verification of Theorem 6.1, and Appendix B the exact searches.

2 Preliminaries

2.1 Kahan's method

Throughout, $F(z) = Q(z,z) + Bz + b$ is a quadratic vector field on $\mathbb{R}^N$, $F'(z) = 2Q(z,\cdot) + B$ is its Jacobian matrix, and $\Phi_h\colon z \mapsto z'$ is Kahan's map (1). We collect standard facts, see [7, 8].

Proposition 2.1. *(a)* $\Phi_h(z) = z + h\big(\mathrm{Id} - \frac{h}{2}F'(z)\big)^{-1}F(z)$*;* Φ_h *is birational and* $\Phi_h^{-1} = \Phi_{-h}$.

(b) If a, b *are affine functions, Kahan's method discretises the quadratic term* $a(z)b(z)$ *as* $\frac{1}{2}[a(z)b(z') + a(z')b(z)]$.

(c) Kahan's method commutes with affine changes of variables: if $z = Tw + c$ *with* T *invertible and* $G(w) = T^{-1}F(Tw + c)$*, then the Kahan map of* G *is* $w \mapsto T^{-1}(\Phi_h(Tw + c) - c)$.

(d) $\det D\Phi_h(z) = \det\big(\mathrm{Id} + \frac{h}{2}F'(z')\big)/\det\big(\mathrm{Id} - \frac{h}{2}F'(z)\big)$.

(e) Every linear first integral of F *is a first integral of* Φ_h.

Definition 2.2. Let I be a function on $\mathbb{R}^N$. A *modified integral* of I is a function I_h, rational in z and smooth in h near $h = 0$, with $I_0 = I$ and $I_h \circ \Phi_h = I_h$. We call I *protected* if it has a modified integral. A measure $\mu = \mathrm{d}z/P(z)$ with P rational is *preserved* if

$$P(\Phi_h(z)) = \det D\Phi_h(z)\, P(z), \tag{2}$$

i.e. P is a discrete Darboux function whose cofactor is the Jacobian determinant [11].

Lemma 2.3. *A protected function is a first integral of* F*, and the protected functions form a vector space.*

Proof. $0 = I_h(\Phi_h(z)) - I_h(z) = h\,\nabla I(z)\cdot F(z) + O(h^2)$. The second claim holds because modified integrals can be added. □

2.2 Quadratic lifts

Definition 2.4. Let $\dot{x} = f(x)$ be an ODE on $\mathbb{R}^n$. A *quadratic lift* consists of a map $\psi\colon \mathbb{R}^n \to \mathbb{R}^m$ and a quadratic vector field F on $\mathbb{R}^{n+m}$ with $F(x,\psi(x)) = \big(f(x), D\psi(x)f(x)\big)$. The *sheet* $\Sigma = \{y = \psi(x)\}$ is invariant under the flow of F, and the flow on Σ projects to the flow of f. The

components of $c = y - \psi(x)$ are the *constraints*. A constraint is *exact* if it is a first integral of F on all of $\mathbb{R}^{n+m}$, and *conditional* if its derivative vanishes only on Σ. The *lifted Kahan method* is Kahan's method applied to F.

The lift is not unique: on Σ the monomial q^2 may be written as q^2 or as y, and more generally any multiple of the constraints may be added to F. By Lemma 2.3, conditional constraints are never protected.

Example 2.5 (the quartic oscillator)**.** For $H = \frac{1}{2}p^2 + \frac{1}{4}q^4$ and $y = q^2$, the lifted field is $\dot q = p$, $\dot p = -qy$, $\dot y = 2qp$. The constraint $c = y - q^2$ is exact, and so is the lifted energy $\tilde H = \frac{1}{2}p^2 + \frac{1}{4}y^2$. Kahan's map conserves $c_h = y - q^2 + \frac{h^2}{4}p^2$ and $\tilde H_h = \tilde H/(1 + \frac{1}{2}h^2q^2)$, and it preserves $\mathrm{d}q\,\mathrm{d}p\,\mathrm{d}y/(1 + \frac{1}{2}h^2q^2)$. Both integrals follow from Section 3.

3 The block theorem

3.1 Statement and proof

Let $P\colon \mathbb{R}^N \to \mathbb{R}^m$ be linear and $w = Pz$ a block of linear coordinates. Suppose that

$$PF(z) = \sum_{k=1}^{r} A_k(z)\, L_k(Pz), \tag{3}$$

where the *speeds* $A_k\colon \mathbb{R}^N \to \mathbb{R}$ are affine and the $L_k(w) = L_k w + \ell_k$ are fixed affine vector fields on $\mathbb{R}^m$. For a function I of degree at most two on $\mathbb{R}^m$, with Hessian $\nabla^2 I$, define

$$(\mathcal{T}_{kl} I)(w) = L_k(w)^\top\, \nabla^2 I\, L_l(w).$$

Lemma 3.1 (crossed identity)**.** *Assume* (3)*, let* $z' = \Phi_h(z)$*,* $w = Pz$*,* $w' = Pz'$*,* $a_k = \frac{h}{2}A_k(z)$ *and* $a'_k = \frac{h}{2}A_k(z')$*. Then*

$$w' - \sum_k a_k L_k(w') = w + \sum_k a'_k L_k(w), \tag{4}$$

and every polynomial I *of degree at most two on* $\mathbb{R}^m$ *that is a first integral of every* L_k *satisfies*

$$I(w') + \frac{1}{2}\sum_{k,l} a_k a_l\, (\mathcal{T}_{kl} I)(w') = I(w) + \frac{1}{2}\sum_{k,l} a'_k a'_l\, (\mathcal{T}_{kl} I)(w). \tag{5}$$

Proof. By Proposition 2.1(b), applied to each product in (3), the w-components of Kahan's equations read $w' - w = \frac{h}{2}\sum_k \big[A_k(z)L_k(w') + A_k(z')L_k(w)\big]$, which is (4). Since I is quadratic, Taylor's formula is exact, and since $\nabla I \cdot L_k = 0$, $I\big(w \pm \sum_k a_k L_k(w)\big) = I(w) + \frac{1}{2}\sum_{k,l} a_k a_l (\mathcal{T}_{kl} I)(w)$. Applying I to both sides of (4) gives (5). □

Theorem 3.2 (block theorem)**.** *Let* V *be a finite-dimensional space of polynomials of degree at most two on* $\mathbb{R}^m$*, each a first integral of every* L_k*, and assume that* $\mathcal{T}_{kl} V \subset V$ *for all* k, l*. Choose a basis* $\mathbf{I} = (I_1, \dots, I_d)$ *of* V *and let* T_{kl} *be the matrices with* $\mathcal{T}_{kl} I_i = \sum_j (T_{kl})_{ij} I_j$*. If the matrices* T_{kl} *commute pairwise, then the vector of functions*

$$\mathbf{I}_h(z) = \mathbf{M}_h(z)^{-1}\, \mathbf{I}(Pz), \qquad \mathbf{M}_h(z) = \mathrm{Id} + \frac{h^2}{8}\sum_{k,l=1}^{r} A_k(z) A_l(z)\, T_{kl}, \tag{6}$$

is conserved by Kahan's map wherever $\mathbf{M}_h$ *is invertible.*

Proof. Apply Lemma 3.1 to each basis element. With $\mathbf{M}(a) = \mathrm{Id} + \frac{1}{2}\sum_{k,l} a_k a_l T_{kl}$ we obtain $\mathbf{M}(a)\mathbf{I}(w') = \mathbf{M}(a')\mathbf{I}(w)$. The matrices $\mathbf{M}(a)$ and $\mathbf{M}(a')$ are polynomials in the commuting T_{kl}, so they commute, and therefore $\mathbf{M}(a')^{-1}\mathbf{I}(w') = \mathbf{M}(a)^{-1}\mathbf{I}(w)$. Since $\mathbf{M}(a) = \mathbf{M}_h(z)$ and $\mathbf{M}(a') = \mathbf{M}_h(z')$, this is the claim. □

The "speeds" cross over in (4): the speed at the old point multiplies the new point and vice versa. If all $a_k = a'_k$, (4) is a Cayley transform and conserves every element of V exactly. The rescaling $\mathbf{M}_h$ is the exact price of sampling the speeds at different ends of the step.

3.2 Consequences

Corollary 3.3 (two variables, [10])**.** *Let $w = Pz \in \mathbb{R}^2$ and $I(w) = \frac{1}{2}a_1 w_1^2 + a_2 w_1 w_2 + \frac{1}{2}a_3 w_2^2 + a_4 w_1 + a_5 w_2$. If $\dot{w} = A(z)J\nabla I(w)$ with A affine and $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, then Kahan's map conserves*

$$I_h = \frac{I + \frac{1}{8}h^2 D_2 A^2}{1 + \frac{1}{4}h^2 D_1 A^2}, \qquad D_1 = a_1 a_3 - a_2^2, \quad D_2 = 2a_2 a_4 a_5 - a_3 a_4^2 - a_1 a_5^2.$$

Proof. Take $V = \operatorname{span}\{I, 1\}$ and $L(w) = J\nabla I(w)$. With $S = \nabla^2 I$ and $s = (a_4, a_5)^\top$ we have $J^\top S J = \operatorname{adj} S$, hence $\mathcal{T} I = (Sw + s)^\top \operatorname{adj}(S)(Sw + s) = 2D_1 I - D_2$ and $\mathcal{T} 1 = 0$. Inverting the 2×2 triangular matrix $\mathbf{M}_h$ gives the formula. □

Corollary 3.4 (affine factors)**.** *Let $I(z) = \frac{1}{2}z^\top S z + s^\top z$ be a first integral of F with $S\,Q(u, v) = 0$ for all u, v, i.e. every linear form in the range of S has an affine time derivative. Then*

$$I_h = I - \frac{h^2}{8} F(z)^\top S\, F(z)$$

is a modified integral.

Proof. Write $S = R^\top \Lambda R$ with R of full row rank and Λ symmetric and invertible, and let $\phi = Rz$ and $\chi = s^\top z$. Since $RQ = 0$, $\dot{\phi} = \alpha(z)$ with α affine, and $\nabla I \cdot F = 0$ gives $\dot{\chi} = -\phi^\top \Lambda \alpha(z)$. The block $w = (\phi, \chi)$ therefore moves as $\dot{w} = \sum_j \alpha_j(z) L_j(w)$ with $L_j(w) = (e_j, -(\Lambda\phi)_j)$. The function $\tilde{I}(w) = \frac{1}{2}\phi^\top \Lambda \phi + \chi$ equals I, is invariant under every L_j, and satisfies $\mathcal{T}_{jk}\tilde{I} = \Lambda_{jk}$ and $\mathcal{T}_{jk} 1 = 0$. All T_{jk} are multiples of one nilpotent matrix, so Theorem 3.2 applies and gives $I_h = I - \frac{h^2}{8}\alpha^\top \Lambda \alpha = I - \frac{h^2}{8} F^\top S F$. □

Example 3.5. For $\dot{x} = y$, $\dot{y} = z$, $\dot{z} = 12xy$ the integral $I = z - 6x^2$ has $S = -12 e_x e_x^\top$, and $\dot{x}$ is linear. Corollary 3.4 gives $I_h = z - 6x^2 + \frac{3}{2}h^2 y^2$, an invariant found by direct computation in [8]. In Example 2.5, the corollary gives c_h.

Example 3.6 (resonant blocks)**.** Let $\mathbb{J}$ be a 2×2 rotation by a right angle and let a block $(u, v) \in \mathbb{R}^2 \times \mathbb{R}^2$ move as $\dot{u} = A(z)(\omega \mathbb{J} u + v)$, $\dot{v} = A(z)\,\omega \mathbb{J} v$. The quadratic invariants of the fixed (Jordan-block) flow include $N = |v|^2$ and $G = u^\top \mathbb{J} v$, and $\mathcal{T} N = 2\omega^2 N$, $\mathcal{T} G = 2\omega^2 G - 2\omega N$. With $\beta = \frac{1}{4}h^2\omega^2 A(z)^2$, Theorem 3.2 gives

$$N_h = \frac{N}{1 + \beta}, \qquad G_h = \frac{G}{1 + \beta} + \frac{\beta N/\omega}{(1 + \beta)^2}.$$

The squared denominator is not produced by Corollaries 3.3 and 3.4: in normal-mode coordinates G is not a sum of two-variable integrals, and its Hessian does not annihilate the quadratic part.

Example 3.7 (non-commuting fields: isotropic angular momenta)**.** For $H = \frac{1}{2}|p|^2 + U(|q|^2)$ with U quadratic, lifted by $y = |q|^2$, the block $w = (q, p) \in \mathbb{R}^{2n}$ moves as $\dot{w} = 1 \cdot L_1(w) + 2U'(y) \cdot L_2(w)$ with $L_1(q, p) = (p, 0)$ and $L_2(q, p) = (0, -q)$. These shear fields do not commute; together with $[L_1, L_2]$ they span $\mathfrak{sl}(2)$. The angular momenta $M_{ab} = q_a p_b - q_b p_a$ are invariant under both, and $\mathcal{T}_{11} M_{ab} = \mathcal{T}_{22} M_{ab} = 0$, $\mathcal{T}_{12} M_{ab} = \mathcal{T}_{21} M_{ab} = M_{ab}$. Theorem 3.2 gives the modified angular momenta

$$M_{ab,h} = \frac{M_{ab}}{1 + \frac{1}{2}h^2 U'(y)}.$$

Remark 3.8. When all coefficient matrices of the Cayley form of Kahan's map commute, [17] shows that z and z' share a "midpoint" at which every compatible quadratic form is conserved. This covers Corollary 3.3, Corollary 3.4 and the single-speed case of Theorem 3.2. In Example 3.7 the coefficient matrices do not commute: their commutator is a non-zero diagonal matrix. Theorem 3.2 only needs the operators $\mathcal{T}_{kl}$ to commute *on* V. For the generalised Suslov systems $\dot{x} = \ell(x)J_0\mathsf{H}x$ of [8], Theorem 3.2 gives one modified integral $I_k/(1 + \frac{1}{4}h^2\ell(x)^2\omega_k^2)$ for each normal-mode action I_k with frequency ω_k.

4 Lift constraints and the shifted sheet

4.1 Which constraints are exact, and which are protected

Proposition 4.1. *Let $c = y - x_ix_j$ be a first integral of a quadratic field F, where y, x_i, x_j are coordinates, and write $F_i = Q_i + (\text{affine})$, $F_j = Q_j + (\text{affine})$ with quadratic forms Q_i, Q_j. Then $x_jQ_i + x_iQ_j = 0$. Consequently:*

(a) if $i = j$, then $Q_i = 0$;

(b) if $i \neq j$, then either $Q_i = Q_j = 0$, or $Q_i = x_iR$ and $Q_j = -x_jR$ for a non-zero linear form R.

Proof. $\nabla c \cdot F = F_y - x_jF_i - x_iF_j = 0$, and F_y is quadratic, so the cubic part $x_jQ_i + x_iQ_j$ vanishes. For $i = j$ this gives $Q_i = 0$. For $i \neq j$ the coordinates x_i and x_j are coprime, so x_i divides Q_i; writing $Q_i = x_iR$ gives $Q_j = -x_jR$. □

In case (b) with $R \neq 0$ the two factors are stretched and squeezed at the same rate, like $(q, p) \mapsto (\mathrm{e}^{\tau}q, \mathrm{e}^{-\tau}p)$. We call this *opposite scaling*.

Theorem 4.2. *If $c = y - x_ix_j$ is a first integral and $\dot{x}_i = F_i$, $\dot{x}_j = F_j$ are affine (cases (a) and (b) with $Q_i = Q_j = 0$), then*

$$c_h = y - x_ix_j + \frac{h^2}{4}F_i(z)F_j(z)$$

is a modified integral.

Proof. Apply Corollary 3.4: $S = -(e_ie_j^{\top} + e_je_i^{\top})$ annihilates Q exactly when $Q_i = Q_j = 0$, and $F^{\top}SF = -2F_iF_j$. □

Corollary 4.3 (mechanical systems with quartic potentials)**.** *Let $H = \frac{1}{2}p^{\top}Gp + V(q)$ with G constant and V a polynomial of degree at most four, and lift with $Y = (y_{ij}) = qq^{\top}$. For every lifted field with $\dot{q} = Gp$ and $\dot{Y} = \dot{q}q^{\top} + q\dot{q}^{\top}$, Kahan's map conserves the matrix $Y - qq^{\top} + \frac{h^2}{4}(Gp)(Gp)^{\top}$.*

In case (b) with $R \neq 0$, Theorem 3.2 does not apply. The block (x_i, x_j, y) moves along the scaling field $(x_i, -x_j, 0)$ and the translation fields $(1, 0, x_j)$, $(0, 1, x_i)$, and the second derivative of c along the scaling field, $2x_ix_j$, is not an invariant of the translations, so closure fails. Such a constraint can still be protected through a *partner*, i.e. by lying in the vector space of Lemma 2.3. Proposition 4.6 gives an example.

4.2 Back-substitution: the shifted sheet

Theorem 4.4. *Let F be a quadratic lift of $\dot{x} = f(x)$ whose constraints $c = y - \psi(x)$ all have modified integrals of the form $c_h = c + h^2r_h(x, y)$ with r_h smooth. For h small, the* shifted sheet *$\Sigma_h = \{c_h = 0\}$ is locally the graph of $y = \psi_h(x) = \psi(x) + O(h^2)$, it is invariant under Φ_h, and the induced map $\varphi_h \colon x \mapsto x'$ is a one-step method of order two for $\dot{x} = f(x)$. If $c_{-h} = c_h$, then φ_h is symmetric, $\varphi_h^{-1} = \varphi_{-h}$. One step costs one linear solve of size $n + m$.*

Proof. The implicit function theorem gives ψ_h, since $\partial c_h/\partial y = \mathrm{Id} + O(h^2)$. Invariance follows from $c_h \circ \Phi_h = c_h$. Since $\Phi_h(z) = z + O(h)$ and $\Phi_h(z+\delta) = \Phi_h(z) + \delta + O(h\delta)$, starting at $(x, \psi_h(x))$ instead of $(x, \psi(x))$ changes the x-component of the step by $O(h^3)$. On Σ the exact flow of F projects to that of f, and Kahan's method has order two, so φ_h has order two. If $c_{-h} = c_h$ then $\Sigma_{-h} = \Sigma_h$, and $\Phi_h^{-1} = \Phi_{-h}$ gives $\varphi_h^{-1} = \varphi_{-h}$. □

Example 4.5. For the quartic oscillator, $\Sigma_h = \{y = q^2 - \frac{h^2}{4}p^2\}$ and substituting into Kahan's equations gives

$$q' = q + \tfrac{h}{2}(p + p'), \qquad p' = p - \tfrac{h}{2}\, qq'(q + q') + \tfrac{h^3}{8}\big(qp'^2 + q'p^2\big).$$

This symmetric map of the plane is *nonlinearly* implicit in (q, p), yet the lifted form computes it with one linear solve. For mechanical lifts (Corollary 4.3) the shifted sheet is explicit, $Y = qq^\top - \frac{h^2}{4}(Gp)(Gp)^\top$.

4.3 One degree of freedom: protection through a partner

Let $x = (q, p) \in \mathbb{R}^2$ and consider $H = U(\varphi(x)) + \kappa(x)$ with $\varphi = \frac{1}{2}x^\top \Phi x$, $\kappa = \frac{1}{2}x^\top K x$ and $U(y) = \frac{1}{2}u_2 y^2 + u_1 y$, $u_2 \neq 0$. Lift with $y = \varphi(x)$ and $\tilde{H} = U(y) + \kappa(x)$. The lifted field is the Nambu field $\dot{z} = \nabla\tilde{H} \times \nabla c$ on $z = (q, p, y)$, with $c = y - \varphi(x)$. This covers all even quartic oscillators ($\varphi = q^2$) and, for $\varphi = qp$, an opposite-scaling constraint.

Proposition 4.6 (essentially [10, Theorem 2])**.** *Let* $\det\Phi \neq 0$*, and let* $\lambda_1 \neq \lambda_2$ *be the roots of* $\det(K - \lambda\Phi) = 0$ *(real, or complex conjugate). Then each pencil member* $I_\lambda = \tilde{H} + \lambda c$ *is a two-variable integral,* $K - \lambda\Phi = \mu_\lambda \ell_\lambda \ell_\lambda^\top$*, and with* $A_\lambda = \ell_\lambda^\top J\Phi x$*,*

$$I_{\lambda,h} = \frac{I_\lambda - \frac{1}{8}h^2\mu_\lambda(u_1 + \lambda)^2 A_\lambda^2}{1 + \frac{1}{4}h^2\mu_\lambda u_2 A_\lambda^2}.$$

Hence $c_h = (I_{\lambda_1,h} - I_{\lambda_2,h})/(\lambda_1 - \lambda_2)$ *and* $\tilde{H}_h$ *are conserved, the measure* $\mathrm{d}z/(D_{\lambda_1}D_{\lambda_2})$ *with* $D_\lambda = 1 + \frac{1}{4}h^2\mu_\lambda u_2 A_\lambda^2$ *is preserved, and the lifted Kahan map is integrable.*

Proof sketch. In the coordinates $(\ell_{\lambda_1}^\top x, y, \ell_{\lambda_2}^\top x)$ both pencil members are two-variable functions sharing y, and $\nabla\tilde{H} \times \nabla c$ is a constant multiple of $\nabla I_{\lambda_1} \times \nabla I_{\lambda_2}$. This is the integrable class of [10, Theorem 2]; the formula for $I_{\lambda,h}$ is Corollary 3.3. □

If $\det\Phi = 0$, one root lies at infinity: φ is then the square of a linear form, c_h is given by Theorem 4.2, $I_{\lambda_1,h}$ by the formula above for the finite root λ_1, and the preserved measure is $\mathrm{d}z/D_{\lambda_1}$. The even quartic oscillators ($\varphi = q^2$) are of this kind.

Remark 4.7 (limits of protection). Protection by Corollary 3.4 reaches one level deep: the *factors* of the constraint must move affinely. In mechanical systems these are the positions, so lifts by products of positions are protected; they represent potentials of degree at most four. A sextic potential needs products of lifted variables, whose velocities are quadratic, and the corresponding constraints are conditional. Rewriting a product by polarisation, $x_i x_j = \frac{1}{2}[(x_i + x_j)^2 - x_i^2 - x_j^2]$, does not help: the condition $SQ = 0$ is invariant under linear changes of variables (Proposition 2.1(c)). Exponentials and single harmonics escape the one-level rule, because their derivatives stay in the span of the lifted variables (Section 5).

5 Invariant measures

5.1 Exponential and trigonometric potentials: the Liouville measure

Consider, on $\mathbb{R}^n \times \mathbb{R}^n$,

$$H = \tfrac{1}{2}p^\top G p + \sum_{k=1}^{K} c_k\, \mathrm{e}^{\alpha_k \cdot q} + \sum_{j=1}^{J} g_j \cos(\beta_j \cdot q), \tag{7}$$

with a constant matrix G and arbitrary vectors $\alpha_k, \beta_j \in \mathbb{R}^n$ (dependent or not). With $u_k = \mathrm{e}^{\alpha_k \cdot q}$ and $(C_j, S_j) = (\cos \beta_j \cdot q, \sin \beta_j \cdot q)$, drop q and consider the quadratic system

$$\dot{p} = -\sum_k c_k u_k \alpha_k + \sum_j g_j S_j \beta_j, \quad \dot{u}_k = (\alpha_k^\top G p)\, u_k, \quad \dot{C}_j = -(\beta_j^\top G p)\, S_j, \quad \dot{S}_j = (\beta_j^\top G p)\, C_j. \tag{8}$$

Theorem 5.1. *Kahan's map of (8) preserves the measure*

$$\mathrm{d}p \prod_{k=1}^{K} \frac{\mathrm{d}u_k}{|u_k|} \prod_{j=1}^{J} \frac{\mathrm{d}C_j\, \mathrm{d}S_j}{C_j^2 + S_j^2}$$

and conserves each modified circle $(C_j^2 + S_j^2)/(1 + \frac{1}{4}h^2(\beta_j^\top G p)^2)$, for any matrix G. If G is symmetric it also conserves $E_h = \frac{1}{2}p^\top G p + \sum_k c_k u_k + \sum_j g_j C_j - \frac{h^2}{8}\dot{p}^\top G \dot{p}$, with $\dot{p}$ given by (8).

Proof. Write $a_k = \alpha_k^\top G p$, $b_j = \beta_j^\top G p$, $\zeta_j = C_j + \mathrm{i}S_j$, and primes for the new values. By Proposition 2.1(b) Kahan's equations are

$$u_k'\big(1 - \tfrac{h}{2}a_k\big) = u_k\big(1 + \tfrac{h}{2}a_k'\big), \qquad \zeta_j'\big(1 - \tfrac{\mathrm{i}h}{2}b_j\big) = \zeta_j\big(1 + \tfrac{\mathrm{i}h}{2}b_j'\big),$$

and $p' - p = \frac{h}{2}\big[-\sum_k c_k \alpha_k (u_k + u_k') + \sum_j g_j \beta_j (S_j + S_j')\big]$. Define intermediate variables $\hat{u}_k = u_k/(1 - \frac{h}{2}a_k)$ and $\hat{\zeta}_j = \zeta_j/(1 - \frac{\mathrm{i}h}{2}b_j)$. Then $u_k + u_k' = \hat{u}_k(2 + \frac{h}{2}\alpha_k^\top G\delta)$ and $S_j + S_j' = 2\hat{S}_j + \frac{h}{2}\hat{C}_j \beta_j^\top G\delta$, where $\delta = p' - p$, so the momentum equation becomes

$$\Big(\mathrm{Id} + \frac{h^2}{4}\Big[\sum_k c_k \hat{u}_k \alpha_k \alpha_k^\top - \sum_j g_j \hat{C}_j \beta_j \beta_j^\top\Big] G\Big)\delta = -h\sum_k c_k \hat{u}_k \alpha_k + h\sum_j g_j \hat{S}_j \beta_j.$$

Thus δ depends only on $(\hat{u}, \hat{\zeta})$. In the coordinates p, $x_k = \log|u_k|$, $\rho_j = \log|\zeta_j|$, $\theta_j = \arg\zeta_j$, one step is the composition of three shears: (x, ρ, θ) is shifted by a function of p; then p is shifted by $\delta(\hat{u}, \hat{\zeta})$; then (x, ρ, θ) is shifted by a function of p'. Each shear preserves $\mathrm{d}p\, \mathrm{d}x\, \mathrm{d}\rho\, \mathrm{d}\theta$, which is the stated measure. The energy is Corollary 3.4: $\dot{p}$ is linear in (u, S), so the Hessian of the lifted energy, which is G on the p-block, annihilates Q. Taking moduli in the ζ_j-equation gives the circles. □

Corollary 5.2. *For the open and periodic Toda chains, rotor and XY chains with nearest-neighbour coupling, and coupled pendula with a non-diagonal mass matrix, the lifted Kahan method is a reversible, volume-preserving, linearly implicit one-step method that conserves a modified energy (and, for Toda, the total momentum).*

The map is not symplectic: for three-particle Toda in the coordinates $(p, \log u)$, the canonical Poisson structure is violated by an amount of order h^3 (about 10^{-2} at $h = 0.2$). Volume preservation by Runge–Kutta methods, of which Kahan's method on quadratic fields is an instance [7], is studied in [41]. The Liouville measure is h-independent, which is unusual for Kahan maps, whose densities typically depend on h. For Lotka–Volterra systems the analogous measure $\mathrm{d}x/\prod_i x_i$ is studied in [37, 38, 17]; Theorem 5.1 concerns the mixed case, with momenta that move affinely and exponential or harmonic variables that move multiplicatively. Volume preservation and reversibility are the two properties required of an integrator in Hybrid Monte Carlo [36], so Theorem 5.1 provides linearly implicit candidates for that purpose.

5.2 One degree of freedom: an area form on the shifted sheet

Write a one-degree-of-freedom potential of degree at most four as a quadratic $\tilde{V}(q, y)$ with $\tilde{V}(q, q^2) = V(q)$. The lift $\dot{q} = p$, $\dot{p} = -\tilde{V}_q - 2q\tilde{V}_y$, $\dot{y} = 2qp$ is the Nambu field $\nabla\tilde{H} \times \nabla c$ on $\mathbb{R}^3$ with $\tilde{H} = \frac{1}{2}p^2 + \tilde{V}$ and $c = y - q^2$. Other lifts of the same potential, which differ by multiples

of $y - q^2$ in $\dot{p}$, are in general not Nambu fields. Kahan maps of three-dimensional quadratic systems have a rich structure [39, 8, 40]. The companion manuscript [17] proves, with computer assistance, that Kahan's map of every three-dimensional Nambu field with quadratic H and C preserves a measure $\mathrm{d}z/P$ with P a polynomial of degree at most four. Our exact searches (Appendix B) independently found such densities for four quartic potentials and for random quadratic Nambu systems, including a closed form valid for all h for $V = \frac{1}{4}q^4 + \frac{1}{3}q^3 + \frac{1}{2}q^2$.

Corollary 5.3. *Assume the result of [17] quoted above. For every quartic potential and the Nambu lift above, the back-substituted method of Theorem 4.4 preserves the area form* $\mathrm{d}q \wedge \mathrm{d}p/P(q, p, q^2 - \frac{h^2}{4}p^2)$.

Proof. On the invariant level set $\{c_h = 0\}$ the preserved measure $\mathrm{d}z/P$ induces the invariant form ω with $\mathrm{d}z/P = \mathrm{d}c_h \wedge \omega$. Since $\partial c_h/\partial y = 1$, its pull-back along the graph $y = q^2 - \frac{h^2}{4}p^2$ is $\mathrm{d}q \wedge \mathrm{d}p/P$. □

For a non-symmetric quartic such as $V = \frac{1}{4}q^4 + \frac{1}{3}q^3 + \frac{1}{2}q^2$, no modified energy is conserved and the degree growth is exponential (Section 7). The back-substituted map is then an area-preserving near-identity map that is not integrable. Area preservation is what KAM-type arguments such as Moser's twist theorem [35] require in order to produce invariant curves that confine orbits, and it is consistent with the bounded energy error observed in practice.

5.3 Measures are rare beyond these classes

Exact searches for polynomial densities (Appendix B) find none for the lifted Kahan maps of two-site φ^4 and Fermi–Pasta–Ulam lattices (degree at most six) and of the two-site discrete nonlinear Schrödinger equation (degree at most four). This agrees with the conjecture of [17] that, for generic quadratic Hamiltonians, Kahan's map of a Lie–Poisson system keeps a measure only for linear systems (two-step nilpotent algebras) and in one degree of freedom; the lifted lattices are affine Lie–Poisson systems on nilpotent algebras of higher step. Among the lifted systems we tested in more than three dimensions, measures came only from symmetry (Section 6) or from exponential and harmonic variables (Theorem 5.1). Other classes have measures for other reasons, for instance Hamiltonian quadratic fields with constant Poisson structure [7] and Lotka–Volterra systems [37, 38].

6 An integrable family in every dimension

Let $n \geq 2$, $U(y) = \frac{1}{2}u_2 y^2 + u_1 y$ with $u_2 \neq 0$, and $H = \frac{1}{2}|p|^2 + U(|q|^2)$ on $\mathbb{R}^n \times \mathbb{R}^n$, a particle in a rotation-invariant anharmonic well. The lift $y = |q|^2$ gives the quadratic field on $\mathbb{R}^{2n+1}$

$$\dot{q} = p, \qquad \dot{p} = -2U'(y)\, q, \qquad \dot{y} = 2\, q \cdot p. \tag{9}$$

Write $Y(z) = 1 + \frac{1}{2}h^2 U'(y)$, $M_{ab} = q_a p_b - q_b p_a$, and $K(z) = \frac{1}{2}|p|^2 + U(y) + \frac{u_1^2}{2u_2} = \frac{1}{2}|p|^2 + \frac{u_2}{2}\big(y + \frac{u_1}{u_2}\big)^2$.

Theorem 6.1. *Kahan's map Φ_h of (9) has the following properties.*

(a) It is $O(n)$-equivariant and (q', p') lies in $\operatorname{span}\{q, p\}$*: explicitly* $\binom{q'}{p'} = N\binom{q}{p}$ *for a 2×2 matrix $N = N(z, z')$ acting on the pair, with* $\det N = Y(z')/Y(z)$*. In particular* $q \wedge q' = h\, q \wedge p / Y(z)$.

(b) It conserves

$$c_h = y - |q|^2 + \frac{h^2}{4}|p|^2, \qquad M_{ab,h} = \frac{M_{ab}}{Y},$$
$$E_h = \frac{K + \frac{1}{2}h^2 u_2 |M_h|^2}{D}, \qquad D = 1 + h^2 u_2 |q|^2 - \frac{1}{2}h^4 u_2 K,$$

where $|M_h|^2 = \sum_{a<b} M_{ab,h}^2$.

(c) It preserves the measure $\mathrm{d}z/P_n$ *with* $P_n = Y^{n-1}D$. *More precisely,* $\det D\Phi_h(z) = \big(Y(z')/Y(z)\big)^{n-1} D(z')/D(z)$.

The linear term u_1 enters only through $\tilde y = y + u_1/u_2$: in the variables $(q, p, \tilde y)$ the field (9) reads $\dot q = p$, $\dot p = -2u_2\tilde y q$, $\dot{\tilde y} = 2q\cdot p$, and $Y = 1 + \frac12 h^2 u_2 \tilde y$, $K = \frac12|p|^2 + \frac12 u_2\tilde y^2$. Since Kahan's method commutes with this affine change of variables, the case $u_1 \neq 0$ is the case $u_1 = 0$ on another level set of c_h. We write $\kappa = h^2u_2$ and use primes for values at z'.

Proof. (a) By Proposition 2.1(b) Kahan's equations are

$$q' - q = \tfrac{h}{2}(p + p'), \qquad p' - p = -\alpha q' - \alpha' q, \qquad y' - y = h(q\cdot p' + q'\cdot p), \tag{10}$$

where $\alpha = hU'(y)$ and $\alpha' = hU'(y')$. The first two say $\left(\begin{smallmatrix} 1 & -h/2 \\ \alpha & 1 \end{smallmatrix}\right)\binom{q'}{p'} = \left(\begin{smallmatrix} 1 & h/2 \\ -\alpha' & 1 \end{smallmatrix}\right)\binom{q}{p}$, where each entry multiplies vectors in $\mathbb{R}^n$. Hence $N = \frac{1}{Y}\left(\begin{smallmatrix} 1-h\alpha'/2 & h \\ -\alpha-\alpha' & 1-h\alpha/2 \end{smallmatrix}\right)$ and $\det N = (1+\frac{h}{2}\alpha')/(1+\frac{h}{2}\alpha) = Y'/Y$. Equivariance holds because (9) is $O(n)$-equivariant (Proposition 2.1(c)).

(b) c_h is Corollary 3.4, since $\dot q = p$ is linear. Since $q'\wedge p' = \det N\, q\wedge p$, we get $M'_{ab}/Y' = M_{ab}/Y$; this is also Example 3.7.

For E_h we may take $u_1 = 0$, so that $\tilde y = y$, by the remark before the proof. Consider the block $w = (p, \tilde y)$. It moves as $\dot w = \sum_k q_k\, L_k(w)$, with the coordinates q_k as speeds and the fixed linear fields $L_k(w) = (-2u_2\tilde y\, e_k,\ 2p_k)$, and $K(w)$ is invariant under every L_k. The closure condition of Theorem 3.2 fails, but Lemma 3.1 still applies. With $a = \frac{h}{2}q$ and $a' = \frac{h}{2}q'$ it gives $K(w') + \frac12\sum_{k,l} a_k a_l (\mathcal{T}_{kl}K)(w') = K(w) + \frac12\sum_{k,l} a'_k a'_l (\mathcal{T}_{kl}K)(w)$, and here $\sum_{k,l} x_k x_l (\mathcal{T}_{kl}K)(w) = 4u_2\big(u_2\tilde y^2|x|^2 + (x\cdot p)^2\big) = 4u_2\big(2K|x|^2 - |x\wedge p|^2\big)$ by Lagrange's identity. Hence

$$K'\big(1+\kappa|q|^2\big) - \tfrac12\kappa|q\wedge p'|^2 = K\big(1+\kappa|q'|^2\big) - \tfrac12\kappa|q'\wedge p|^2. \tag{11}$$

By (a) and the first equation in (10), $q\wedge p' = \frac{2}{h}\, q\wedge q' - q\wedge p = (2-Y)M_h$ and $q'\wedge p = \frac{2}{h}\, q\wedge q' - q'\wedge p' = (2-Y')M_h$. Conservation of c_h gives $|q|^2 = \tilde y + \frac{h^2}{2}K - \frac{h^2}{4}u_2\tilde y^2 - g$ and the same expression with primes, with the same constant g. Substitute these into (11); the terms in KK' cancel. With $\lambda = \frac12\kappa|M_h|^2$, $s = 2 - Y = 1 - \frac12\kappa\tilde y$ and $\mu = 2 - \kappa g$, what remains is $K'(\mu - s^2) - \lambda s^2 = K(\mu - s'^2) - \lambda s'^2$, that is,

$$(K' + \lambda)(\mu - s^2) = (K + \lambda)(\mu - s'^2).$$

Finally $\mu - s^2 = 1 + \kappa\tilde y - \frac14\kappa^2\tilde y^2 - \kappa g = D$ and $\mu - s'^2 = D'$, so $E'_h = E_h$.

(c) Put $\xi = q - \frac{h}{2}p$, $\eta = q + \frac{h}{2}p$ and $\eta' = q' + \frac{h}{2}p'$. The first equation in (10) says $\eta = q' - \frac{h}{2}p'$, so $q = \frac12(\xi + \eta)$ and $q' = \frac12(\eta + \eta')$. Let $A = \xi\cdot\eta$, $B = \eta\cdot\eta'$ and $C = |\eta|^2$. In these variables the third equation in (10) reads $y' - y = B - A$, and the second one reads $Y\eta' + Y'\xi = (4 - Y - Y')\,\eta$. Hence

$$Y' - Y = \tfrac12\kappa(B - A), \qquad YB + Y'A = (4 - Y - Y')\,C. \tag{12}$$

Moreover $D = (2-Y)Y + \kappa A$ and $D' = (2-Y')Y' + \kappa B$, because $1 - \frac14\kappa^2\tilde y^2 = (2-Y)Y$ and $|q|^2 - \frac{h^2}{4}|p|^2 = \xi\cdot\eta$. A Schur-complement computation in the block form of $\mathrm{Id} \mp \frac{h}{2}F'$ gives

$$\begin{aligned} \det\big(\mathrm{Id} - \tfrac{h}{2}F'(z)\big) &= Y^{n-1}\big(Y + \kappa\, q\cdot\eta\big) = Y^{n-1}\big(Y + \tfrac12\kappa(A + C)\big), \\ \det\big(\mathrm{Id} + \tfrac{h}{2}F'(z')\big) &= Y'^{n-1}\big(Y' + \kappa\, q'\cdot\eta\big) = Y'^{n-1}\big(Y' + \tfrac12\kappa(B + C)\big). \end{aligned}$$

Using (12), a short computation shows

$$2D = (4 - Y - Y')\big(Y + \tfrac12\kappa(A + C)\big), \qquad 2D' = (4 - Y - Y')\big(Y' + \tfrac12\kappa(B + C)\big).$$

For the first identity, expand the right-hand side, replace $(4 - Y - Y')C$ by $YB + Y'A$, and then $\frac12\kappa(B - A)$ by $Y' - Y$; the second follows by symmetry. Where $4 - Y - Y' \neq 0$, dividing the two identities and using Proposition 2.1(d) gives $\det D\Phi_h = (Y'/Y)^{n-1}D'/D$, which is equivalent to $P_n(z') = \det D\Phi_h\, P_n(z)$. Where $4 - Y - Y' = 0$, both D and D' vanish and the measure identity holds trivially. □

The integrals and the measure were first found by exact searches for Darboux polynomials (Appendix B). The proofs explain their form. The angular momenta come from the block theorem with non-commuting fields. The energy comes from a block $(p, \tilde{y})$ that fails the closure condition only by the term $|q \wedge p|^2$, which the conserved discrete angular momentum absorbs. The measure density is Y^{n-1} times the denominator of E_h. For $M = 0$, E_h is a function of the two-variable integral $K/(1+\kappa|q|^2)$ of Corollary 3.3. All identities were also verified independently by computer algebra (Appendix A).

Corollary 6.2 (integrability)**.** *For $n = 2$ the integrals c_h, M_h, E_h are functionally independent. Let L be a compact connected component of a regular level set with $M_h \neq 0$ on which $\Phi_h^{\pm 1}$ are defined and $P_2 \neq 0$, and assume $\Phi_h(L) = L$ and that the induced circle map below preserves orientation (both hold for small h). Then L is a 2-torus and $\Phi_h|_L$ is smoothly conjugate to a skew rotation*

$$(\theta, \phi) \mapsto (\theta + \rho,\ \phi + g(\theta)), \qquad (\theta, \phi) \in \mathbb{T}^2.$$

If ρ is Diophantine, $\Phi_h|_L$ is conjugate to a translation of $\mathbb{T}^2$, i.e. the motion is quasi-periodic. For $n > 2$ the same holds on the level sets of c_h, E_h and all $M_{ab,h}$, which comprise $2n-1$ independent integrals on $\mathbb{R}^{2n+1}$.

Proof. A 3×3 minor of the Jacobian of (c_h, M_h, E_h) is a non-zero rational function (Appendix A), which gives functional independence. L is a compact, orientable 2-manifold, being a regular level set in $\mathbb{R}^5$. Rotations of the pair (q, p) preserve the integrals, so the connected group $SO(2)$ maps L to itself; it acts freely because $M \neq 0$. A compact orientable surface with a free circle action is a torus, and $L/SO(2)$ is a circle. Φ_h commutes with the action, so it induces a homeomorphism f of $L/SO(2)$. The measure $\mathrm{d}z/P_2$ induces on L the invariant density $\sigma = (\mathrm{d}z/P_2)/(\mathrm{d}c_h \wedge \mathrm{d}M_h \wedge \mathrm{d}E_h)$, which is smooth, nowhere zero and rotation invariant, so $|\sigma|$ descends to a smooth positive f-invariant density on the circle. Its distribution function conjugates f to a rigid rotation by ρ. Since $M \neq 0$, $q \neq 0$ on L, so $\phi = \arg q$ (in the plane of motion) is a coordinate on which $SO(2)$ acts by translation; together with a coordinate θ on the circle $L/SO(2)$ in which f is the rotation, it gives coordinates (θ, ϕ) on L. Since Φ_h commutes with the translations of ϕ, it has the stated form, with $g(\theta) = \arg q' - \arg q$. By (a), $q \wedge q' = hM_h \neq 0$, so this increment stays in $(0, \pi)$ or in $(-\pi, 0)$, and g is a smooth real-valued function. For Diophantine ρ the cohomological equation $\gamma(\theta + \rho) - \gamma(\theta) = g(\theta) - \bar{g}$ has a smooth solution, and $\phi \mapsto \phi - \gamma(\theta)$ conjugates to the translation by $(\rho, \bar{g})$. For $n > 2$, fixing all $M_{ab,h}$ fixes the oriented plane Π (the bivector M_h is decomposable, giving $2n-3$ independent components) and reduces to $n = 2$. □

Proposition 6.3 (bounded orbits)**.** *Let $u_2 > 0$. If $D(z_0) > 0$, for instance if $h^4 u_2 K(z_0) < 2$, then the orbit of z_0 under Φ_h is bounded, as long as it is defined.*

Proof. Let $e = E_h(z_0) \geq 0$ and $c = c_h(z_0)$. Along the orbit $K + \frac{1}{2}\kappa|M_h|^2 = eD$, hence $K\big(1 + \frac{1}{2}eh^4u_2\big) \leq e\big(1 + \kappa|q|^2\big)$, that is, $K \leq e_*\big(1 + \kappa|q|^2\big)$ with $e_* = e/(1 + \frac{1}{2}eh^4u_2)$ and $e_* h^4 u_2 < 2$. Put $s = \frac{h^2}{4}|p|^2 \geq 0$ and $t = |q|^2 - s$, so that $y = c + t$. The inequality becomes

$$s\Big(\frac{2}{h^2} - e_* h^2 u_2\Big) + \frac{u_2}{2}\Big(t + c + \frac{u_1}{u_2}\Big)^2 - e_* h^2 u_2\, t \leq e_*.$$

The coefficient of s is positive, so the inequality bounds t (a quadratic with positive leading coefficient) and then s. Hence $|q|^2 = t + s$, $|p|^2$ and y stay bounded. □

Remark 6.4 (an $O(n)$ explicit form). Eliminating q', p' from Kahan's equations leaves one scalar linear equation for y'. With $\alpha = hU'(y)$, $Y = 1 + \frac{h}{2}\alpha$ and $\eta = q + \frac{h}{2}p$,

$$y' = \frac{yY + h\big(2p - (\alpha + hu_1)q\big)\cdot\eta}{Y + \kappa\, q\cdot\eta},$$

$$q' = \frac{(1 - \frac{h}{2}\alpha')q + hp}{Y}, \qquad p' = \frac{(1 - \frac{h}{2}\alpha)p - (\alpha + \alpha')q}{Y},$$

Table 1: Degree growth $d_1, d_2, \ldots$ of lifted Kahan maps (exact, modulo a prime, along a random line). Polynomial (here linear or quadratic) growth indicates integrability, exponential growth chaos. "Source" names the origin of the continuous system's integrability. Lifts use $y = |q|^2$ for isotropic systems, $y_k = q_k^2$ (or the products noted) for quartic potentials, and exponential or harmonic variables otherwise.

Lifted system	$d_1, d_2, \ldots$	Source of integrability
Integrable lifted maps		
quartic oscillator ($\mathbb{R}^3$)	$3, 9, 19, 33, 51, 73, 99$	one degree of freedom
isotropic quartic, $n = 2, 3, 4$	$4, 12, 25, 44, 68, 97, 132$	rotations, one d.o.f.
isotropic quartic plus harmonic term	$4, 12, 25, 44, 68, 97$	rotations, one d.o.f.
isotropic quartic, magnetic field, $n = 2$	$4, 13, 28, 49, 76, 109$	rotations, one d.o.f.
decoupled $q_1^4 + q_2^4$	$5, 17, 37, 65, 101, 145$	decoupling
1:6:1 quartic, lift in $q_1 \pm q_2$	$5, 17, 37, 65, 101, 145$	decoupling after rotation
two-particle Toda	$2, 4, 6, 8, 10, 12$	translation, one d.o.f.
two coupled rotors	$3, 9, 19, 33, 51, 73$	translation, one d.o.f.
Non-integrable lifted maps		
1:6:1 quartic, lift $y_k = q_k^2$	$5, 25, 113, > 400$	same system, unadapted lift
quartic 1:6:8	$5, 25, 113, > 400$	hidden integral, quartic in momenta
quartic 1:12:16	$5, 25, 113, > 400$	parabolic separation
Garnier, $a_1 \neq a_2$	$4, 16, 58, 208, > 400$	elliptic separation
open Toda, three particles	$3, 9, 23, 57, 139, 337, 815$	Lax pair
anisotropic $(q_1^2 + 2q_2^2)^2$	$4, 16, 58, 208, 742$	none (not integrable)
sextic isotropic	$5, 16, 43, 114, 305, > 400$	lift not fully protected
non-symmetric quartic, one d.o.f.	$3, 9, 23, 57, 139, 337, 815$	energy not protected
prolonged Euler top ($v = \dot{x}$)	$4, 16, 52, 160, > 400$	integrals not protected

with $\alpha' = hU'(y')$. A step therefore costs $O(n)$ operations, like one Störmer–Verlet step, and no linear system has to be solved. The denominators vanish exactly where $\det(\mathrm{Id} - \frac{h}{2}F') = Y^{n-1}(Y + \kappa\, q\cdot\eta)$ does. Two further identities make the link with discrete Newtonian equations [27]. Along an orbit (q_k, p_k, y_k),

$$q_k \wedge q_{k+1} = h\, M_h, \qquad q_{k+1} + q_{k-1} = \lambda_k\, q_k, \quad \lambda_k = \frac{2 - \frac{1}{2}h^2\big(U'(y_{k+1}) + U'(y_{k-1})\big)}{1 + \frac{1}{2}h^2 U'(y_k)}.$$

The first identity exhibits M_h as the discrete angular momentum, and the second is a "central force" recurrence in which the force is evaluated through the auxiliary sequence y_k. With the initial value $y_0 = |q_0|^2 - \frac{h^2}{4}|p_0|^2$ on the shifted sheet, (q_k, p_k) is a symmetric, second-order, integrable discretisation of the original system on $\mathbb{R}^{2n}$ (Theorem 4.4).

7 Which integrable systems survive lifting?

Proposition 4.6 and Theorem 6.1 produce integrable maps from systems whose integrability has a simple origin: one degree of freedom, possibly after reduction by a linear symmetry. To see how far this goes, we computed the degree growth of iterates for nineteen lifted Kahan maps. Kahan maps are birational; restricted to a generic line, the k-th iterate has some degree d_k, and zero algebraic entropy $\lim_{k\to\infty} \frac{1}{k}\log d_k = 0$, in practice polynomial growth of d_k, is the standard indicator of integrability [31, 32]. We computed d_k exactly, modulo a large prime (Appendix B). Table 1 and Fig. 1 summarise the results.

Three observations stand out. First, the isotropic family stays integrable with a harmonic term, in dimensions three and four, and in a uniform magnetic field. Second, in our tests every lifted system whose continuous integrability rests on a hidden symmetry is chaotic after lifting: three-particle Toda, the Garnier system [34], and the quartic potentials 1:6:8 and 1:12:16 [33].

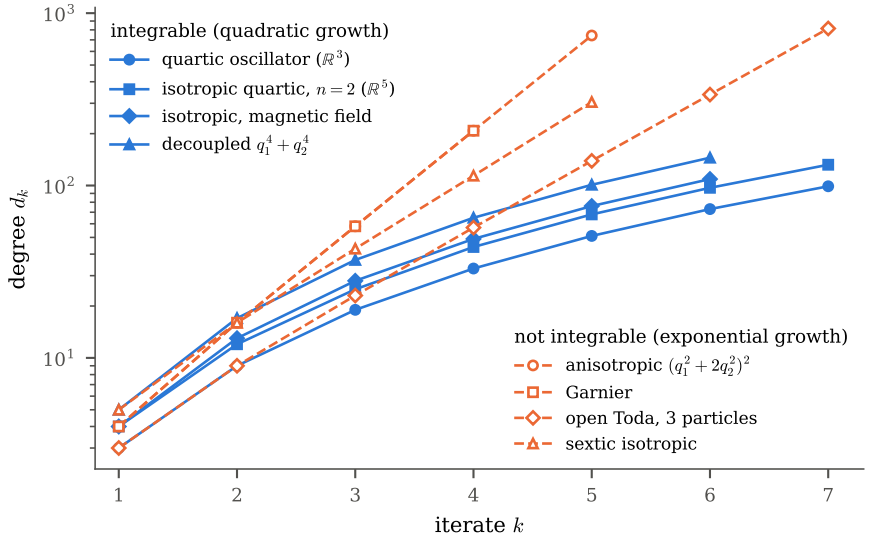


Figure 1: Degree growth of representative lifted Kahan maps (logarithmic scale). Integrable maps (blue, solid) grow quadratically; non-integrable ones (orange, dashed) grow exponentially. The Garnier sequence coincides with the anisotropic one for $k \leq 4$.

Third, the separable 1:6:1 potential $q_1^4 + 6q_1^2q_2^2 + q_2^4 = \frac{1}{2}[(q_1+q_2)^4 + (q_1-q_2)^4]$ gives an integrable map with a lift in the separating coordinates $q_1 \pm q_2$ and a chaotic map with the lift $y_k = q_k^2$. Kahan's method commutes with linear changes of variables, but these two lifts are not related by one, so the lift must respect the linear structure that makes the system integrable.

Conjecture 7.1. *Let $\dot{x} = f(x)$ be integrable because of a group of linear symmetries together with a reduction to one degree of freedom, and let F be a quadratic lift that is equivariant under the symmetries, adapted to the reduction and has only protected constraints. If the reduced one-degree-of-freedom problem lies in a class on which Kahan's method is integrable (Proposition 4.6, or the single-exponential and single-harmonic cases of Theorem 5.1), then the lifted Kahan map is integrable.*

The heuristic reason is that every mechanism producing exact modified integrals here is linear-algebraic: Kahan's method commutes with linear maps, and the modified integrals are built from quadratic invariants of linear flows and their speeds. A linear symmetry therefore passes to the map, while hidden integrals (Toda's $\operatorname{tr} L^3$, Garnier's Uhlenbeck integrals, the quartic integrals of 1:6:8) are not invariants of linear flows and nothing protects them. This is a statement about *lifted* systems only: the genuinely quadratic Clebsch and $\mathfrak{so}(4)$ Manakov tops keep hidden quadratic integrals under Kahan's method [5, 6].

The magnetic case shows that Theorem 3.2 is sufficient but not necessary for protection. In velocity variables, $\dot{q} = v$, $\dot{v} = -2U'(y)q + B\,\mathbb{J}v$, $\dot{y} = 2q \cdot v$, the canonical angular momentum fails the closure condition of Theorem 3.2, yet the degree growth is quadratic. For $U(y) = \frac{1}{4}y^2$, $B = 1$ and $h = \frac{1}{5}$, an exact search finds two invariant densities $\mathrm{d}z/P_1$ and $\mathrm{d}z/P_2$ with polynomials P_1, P_2 of degrees three and four. Their ratio is a rational integral, conserved to round-off, that mixes $q \wedge v$ with energy-like terms. No mechanism in this paper predicts it.

8 Numerical experiments

We test the isotropic family as an integrator for a stiff anharmonic oscillator,

$$H = \tfrac{1}{2}|p|^2 + \tfrac{1}{2}\omega^2|q|^2 + \tfrac{1}{4}|q|^4, \qquad q, p \in \mathbb{R}^3, \quad \omega = 10,$$

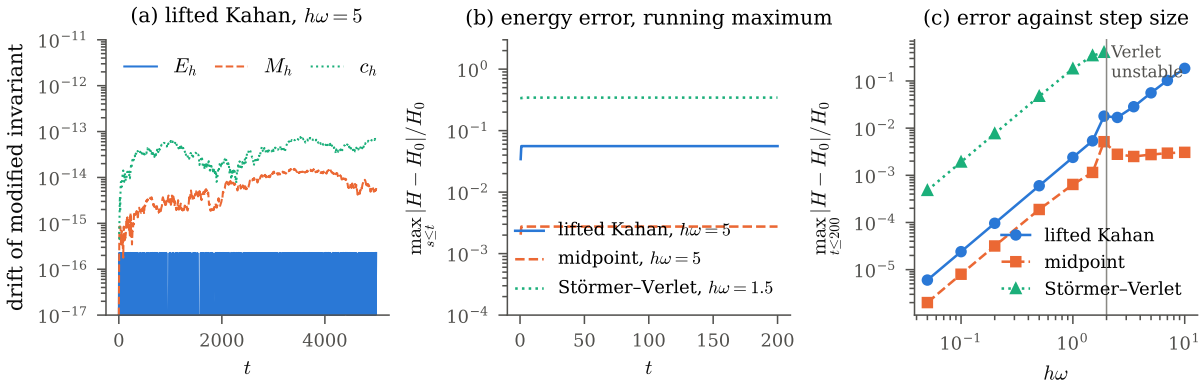


Figure 2: Stiff isotropic oscillator in $\mathbb{R}^3$, $\omega = 10$. (a) Lifted Kahan at $h\omega = 5$: drift of the modified invariants over 10^4 steps (relative for E_h and M_h, absolute for c_h). (b) Running maximum of the relative energy error. (c) Maximal relative energy error on $[0, 200]$ against $h\omega$; Störmer–Verlet is unstable for $h\omega > 2$.

i.e. $u_2 = \frac{1}{2}$ and $u_1 = \frac{1}{2}\omega^2$, with $q_0 = (0.8, 0.1, -0.3)$, $p_0 = (0, 3, 1)$ and y_0 on the shifted sheet. The stiff harmonic term sits in the linear part of the lift, which Kahan's method treats like the midpoint rule. We compare with the implicit midpoint rule on (q, p), solved by Newton's method to a tolerance of 10^{-14}, and with the Störmer–Verlet method. Figure 2 shows the results.

- *Exact invariants.* At $h\omega = 5$ the modified invariants E_h, M_h and c_h drift by at most 2.2×10^{-16}, 1.6×10^{-14} and 7.7×10^{-14} over 10^4 steps, i.e. by round-off. The true energy error stays bounded at 5.6×10^{-2}.

- *Stability.* Störmer–Verlet is unstable for $h\omega > 2$. For the initial value above, the lifted Kahan map remained bounded for all tested step sizes up to $h\omega = 10$, and so did 20 random orbits with $|q_0| \le 3$ and $|p_0| = \omega|q_0|$ for $h\omega \le 5$. Proposition 6.3 guarantees bounded orbits only when $D(z_0) > 0$, which here means $h\omega$ below about 2, because the constant $u_1^2/(2u_2) = \omega^4/4$ dominates K. Beyond that the level sets of E_h need not be compact, and for large initial data some orbits do escape: one of 20 random orbits with $|q_0| = 3$ escaped at $h\omega = 10$, and an independent test with other initial data found more such escapes. Stability beyond $h\omega \approx 2$ is therefore observed at moderate amplitudes, not guaranteed.

- *Accuracy and cost.* At equal step size the energy error of the lifted Kahan map is about three times that of the midpoint rule for $h\omega \le 1$, and it grows to 0.185 at $h\omega = 10$, where the midpoint error saturates at 3×10^{-3}. The midpoint rule needed three to four Newton iterations per step, each with a 6×6 linear solve, whereas the lifted Kahan step (Remark 6.4) costs about as much as a Störmer–Verlet step.

So the lifted Kahan map is an explicit-cost, symmetric, integrable integrator whose qualitative long-time behaviour (invariant tori, exact modified energy and angular momenta) is correct, at moderate amplitudes, at step sizes where the standard explicit method fails. It is less accurate than the fully implicit midpoint rule at large $h\omega$. For the exponential and trigonometric class of Theorem 5.1, stiffness that enters through the lifted nonlinear variables limits the step size: for a stiff rotor chain the lifted Kahan method conserved its modified energy and stayed stable up to time 600 for $h\omega \le 2$, and blew up beyond.

9 Conclusions and open problems

Quadratic lifts let Kahan's method act on polynomial, exponential and trigonometric problems. Exact modified integrals survive through a few linear-algebraic mechanisms, collected in

Theorem 3.2: affinely moving factors, two-variable integrals, and blocks that move along fixed linear flows at affine speeds, possibly several and non-commuting. These mechanisms protect the lift constraints of quartic potentials, give the Liouville measure for exponential and harmonic potentials, and make the lifted isotropic quartic oscillator integrable in every dimension, with an $O(n)$ explicit form. Open problems:

1. *A block theorem without closure.* The proof of Theorem 6.1(b) uses a block whose closure defect is absorbed by another conserved quantity. A general version of this mechanism, or a reduction theory for Kahan maps with linear symmetries, might explain the magnetic case and prove Conjecture 7.1.

2. *The magnetic case.* Explain the rational integral of the magnetic isotropic map, and test dimensions $n \geq 3$.

3. *Explicit solutions.* The invariant tori of Corollary 6.2 should admit a parametrisation by elliptic functions, as for planar Kahan maps [13].

4. *Higher order.* Modified integrals depend on h, so compositions of Kahan steps with different step sizes conserve none of them exactly. Is there a lift or a composition that keeps a common invariant?

5. *Applications.* Volume-preserving, reversible, linearly implicit integrators for exponential and rotor lattices (Corollary 5.2) are natural candidates for Hybrid Monte Carlo [36]; their acceptance rates remain to be tested.

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[To be completed.] Parts of the symbolic and numerical computations and of the drafting of this manuscript were carried out with the assistance of an AI system; all results were checked by the author.

Data availability

The scripts that reproduce every computation and figure, including the computer-algebra verification of Theorem 6.1, are available at [repository URL to be added].

A Computer-algebra verification of Theorem 6.1

The proofs of Theorem 6.1 in Section 6 are complete without computer assistance. As an independent check, three SymPy 1.14 scripts in the folder `proofs/` of the repository verify the statements as exact identities between rational functions, with h, u_1 and u_2 symbolic.

1. `isotropic_proof.py` (about 8 minutes) first checks that the closed form of Remark 6.4 satisfies Kahan's equations identically for $n = 2, 3$, so that it *is* Kahan's map. It then checks the conservation of c_h, M_h and E_h and the measure identity $P_n(z')\det(\mathrm{Id} - \frac{h}{2}F'(z)) = \det(\mathrm{Id} + \frac{h}{2}F'(z'))P_n(z)$ for $n = 2$ and $n = 3$. Since the map and all functions involved are rotation invariant, it suffices to check these on the slices $q = (q_1, 0)$ and $q = (q_1, 0, 0)$, $p = (p_1, p_2, 0)$; the image point is not restricted. Finally it computes a 3×3 minor of the Jacobian of (c_h, M_h, E_h) at $q = (1, 0)$, $p = (0, 1)$, $y = \frac{1}{3}$, which is a non-zero rational function of (h, u_1, u_2); this gives the functional independence used in Corollary 6.2.

2. `isotropic_hand_proof_check.py` verifies each displayed step of the hand proofs, for $n = 3$: (11), the expressions for $q \wedge p'$ and $q' \wedge p$, the relations (12), the two determinants, and the two identities for $2D$ and $2D'$.

3. `isotropic_proof_adjugate.py` computes the step from the adjugate formula $z' = z + h\,\mathrm{adj}(\mathcal{M})F(z)/\det\mathcal{M}$, $\mathcal{M} = \mathrm{Id} - \frac{h}{2}F'(z)$, instead of the closed form, and repeats the checks for $n = 2$. It is slower: the energy takes about 7 minutes, and the measure identity about 10 minutes for $u_2 = \frac{1}{2}$, $u_1 = 0$.

Independent floating-point and exact rational checks at random points, for $n = 2, 3, 4$, are also in the repository.

B Exact searches and degree growth

Darboux polynomials and densities. A polynomial P of degree at most d with $P(\Phi_h(z)) = \lambda(z)P(z)$ for a given cofactor λ is found by solving a linear system for its coefficients. We impose the identity at many random points, in arithmetic modulo a large prime p and with a random residue for h, following the discrete-Darboux method of [11]. With the cofactor $\lambda = \det D\Phi_h$ this finds invariant densities $\mathrm{d}z/P$. Since h is a random residue, a solution found this way exists for generic h, and "no solution up to degree d" is an exact negative result for that degree, barring an unlucky choice of prime. All positive results were then confirmed over the rationals or with symbolic h. The modified energy E_h of Theorem 6.1 was found as the ratio of two Darboux polynomials with the cofactor $(K'/K)(Y'/Y)^2$, where K and Y are the factors of the measure.

Degree growth. To compute d_k we restrict the k-th iterate to a random line $z = a + bt$ over $\mathbb{F}_p$ and determine the degree of the resulting rational function of t from the nullity of a Padé-type linear system. The iterates are evaluated numerically modulo p rather than symbolically, so degrees of several hundred are within reach. Values above about 400 were not resolved and are reported as "> 400".

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